

Notes: Baker, Gibbons & Murphy (QJE, 2002)

Relational Contracts and the Theory of the Firm

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1 Big picture

- **Question.** How does the boundary of the firm affect the relational contracts that parties can sustain when performance is observable to the parties but not verifiable to courts?
- **Central mechanism.** Asset ownership changes what happens after reneging. This changes the short-run temptation to renege and therefore changes which informal agreements are self-enforcing.
- **Key distinction.** Under relational employment, the downstream party owns the asset and can take the good after reneging. Under relational outsourcing, the upstream party owns the asset and retains bargaining recourse through the spot market.
- **Main proposition.** The same bonus contract can induce the same actions and total surplus under either ownership structure, but the total reneging temptation differs across ownership structures.
- **Organizational implication.** Integration is not just about residual control rights in a static incomplete-contracting model. It is also a tool for supporting, weakening, or reshaping relational contracts.
- **Selective intervention implication.** A firm cannot simply bring a market transaction inside and mimic spot-market incentives, because the required relational payments would create reneging temptations too large to sustain.
- **Comparative implications.** Vertical integration can be efficient when outside supply values vary widely; performance payments tend to be lower inside firms; the optimal boundary can depend on the discount rate and payoff levels; and the existence of a mediocre spot market can block otherwise efficient firms.
- **Takeaway.** Firm boundaries and informal contracts are jointly determined. A theory of the firm must explain not only who owns assets but also how ownership affects the credibility of informal promises.

2 Incremental contribution of the paper

- **Combines two literatures that were usually separate.** The paper integrates property-rights theories of asset ownership with repeated-game theories of relational contracts.
- **Compares relationships within and between firms symmetrically.** Prior work often studied relational contracts inside firms or between firms separately. This paper places both in the same model.
- **Endogenizes reneging temptations.** Instead of taking relational capital or hold-up risk as reduced-form primitives, the paper derives reneging temptations from asset ownership and spot-market fallback payoffs.
- **Changes the interpretation of integration.** Integration can be chosen because it makes a valuable relational contract feasible, not because it directly improves formal contracting or information.
- **Formalizes the impossibility of selective intervention.** The paper gives a repeated-game foundation for Williamson's argument that firms cannot costlessly mimic markets and intervene only when markets fail.
- **Generates testable organizational predictions.** The model predicts lower-powered performance payments inside firms, integration under volatile supply prices, and ownership changes as discount rates or outside-option levels change.
- **Gives a role to management.** Managers matter because they design, communicate, implement, and revise relational contracts that courts cannot enforce.

3 Model primitives and measurement

3.1 Parties, asset, and ownership

There is an upstream party, a downstream party, and one asset. The upstream party uses the asset to produce a good each period. If the upstream party owns the asset, the transaction is nonintegrated and is called outsourcing. If the downstream party owns the asset, the transaction is integrated and is called employment. Ownership of the asset conveys ownership of the good produced by the asset.

Interpretation. Asset ownership is the formal organizational variable. It determines who can walk away with the good after reneging.

3.2 Values of the good

The downstream value is either high or low, Q_H or Q_L . The alternative-use value is either high or low, P_H or P_L . The ranking is $P_L < P_H < Q_L < Q_H$, so efficient use is always downstream. The upstream party's actions affect the probabilities of high downstream and alternative-use values.

Interpretation. The model allows an action to improve internal value, external value, or both. This is what makes ownership and relational contracts interact.

3.3 Actions, cost, and first best

The upstream party chooses a vector of costly actions $a = (a_1, \dots, a_n)$ at cost $c(a)$. The high downstream probability is $q(a)$; the high alternative-use probability is $p(a)$. The efficient benchmark maximizes expected downstream value net of action cost.

Interpretation. The first best ignores incentive and reneging constraints. It is the target that relational contracts try to approximate.

3.4 Observable but nonverifiable outcomes

Q and P are observed by the parties after production but are not verifiable by courts. Relational contracts can condition on Q and P because the parties observe them. Self-enforcement requires that neither party wants to renege after the realized state is observed.

Interpretation. The model separates observability from verifiability. Relational contracts use information that formal contracts cannot use.

3.5 Governance regimes

The four regimes are spot outsourcing, spot employment, relational outsourcing, and relational employment.

Interpretation. The paper's main comparison is the bottom row of Figure I: relational outsourcing versus relational employment.

4 Models and theoretical specifications

4.1 First-best benchmark

$$\begin{aligned}\Delta Q &= Q_H - Q_L, & \Delta P &= P_H - P_L, \\ S^{FB} &= Q_L + q(a^*)\Delta Q - c(a^*), & a^* &\in \arg \max_a \{Q_L + q(a)\Delta Q - c(a)\}.\end{aligned}$$

Interpretation. The first-best action maximizes downstream surplus because the good is always more valuable downstream than in its alternative use. The alternative-use value affects incentives and reneging, not the efficient use of the good.

4.2 Spot outsourcing

$$\begin{aligned}\text{spot price after state } (Q_i, P_j) &= \frac{1}{2}(Q_i + P_j), \\ a^{SO} &\in \arg \max_a \left\{ \frac{1}{2}[Q_L + q(a)\Delta Q] + \frac{1}{2}[P_L + p(a)\Delta P] - c(a) \right\}, \\ S^{SO} &= Q_L + q(a^{SO})\Delta Q - c(a^{SO}).\end{aligned}$$

Interpretation. Spot outsourcing gives upstream bargaining power because upstream owns the good. This gives upstream incentives to raise the alternative-use value P , even when that does not improve downstream surplus.

4.3 Spot employment

$$a^{SE} = 0, \quad S^{SE} = Q_L.$$

Interpretation. Under spot employment, downstream owns the good and can take it without paying a bonus. Anticipating this, upstream takes no costly noncontractible actions.

4.4 Relational employment

$$\begin{aligned}S^{RE} &= Q_L + q(a^{RE})\Delta Q - c(a^{RE}), \\ \max_{ij} b_{ij} - \min_{ij} b_{ij} &\leq \frac{1}{r} [S^{RE} - \max\{S^{SE}, S^{SO}\}].\end{aligned}$$

Interpretation. The left side is the maximum total temptation to renege on promised payments. The right side is the discounted value of preserving the relationship relative to the best spot fallback.

4.5 Relational outsourcing

$$\begin{aligned}S^{RO} &= Q_L + q(a^{RO})\Delta Q - c(a^{RO}), \\ \max_{ij} \left[b_{ij} - \frac{1}{2}(Q_i + P_j) \right] - \min_{ij} \left[b_{ij} - \frac{1}{2}(Q_i + P_j) \right] &\leq \frac{1}{r} [S^{RO} - \max\{S^{SO}, S^{SE}\}].\end{aligned}$$

Interpretation. For a given bonus schedule, actions and surplus can match relational employment. The difference is the reneging temptation: under outsourcing, reneging leads to bargaining around the spot price.

4.6 Main proposition

The same relational contract produces the same actions and total surplus under either ownership structure, but the maximum total reneging temptation is different under relational employment and relational outsourcing.

Interpretation. Ownership changes what informal agreements are credible, even when it does not change information, preferences, technology, or formal contractibility.

4.7 Linear two-task example

$$q(a) = q_1 a_1 + q_2 a_2, \quad p(a) = p_1 a_1 + p_2 a_2, \quad c(a) = \frac{1}{2} a_1^2 + \frac{1}{2} a_2^2.$$

$$a_i^{FB} = q_i \Delta Q, \quad a_i^{SO} = \frac{1}{2} q_i \Delta Q + \frac{1}{2} p_i \Delta P, \quad a_i^{SE} = 0.$$

For relational bonuses Δb on Q and $\Delta \beta$ on P ,

$$a_i^R = q_i \Delta b + p_i \Delta \beta.$$

Interpretation. The two tasks can raise downstream value, alternative-use value, or both. In the unproductive-multitasking case, actions that raise P can waste resources but improve bargaining position under outsourcing.

4.8 Simplified relational constraints and first-best feasibility

$$|\Delta b| + |\Delta \beta| \leq \frac{1}{r} [S^{RE} - \max\{S^{SO}, S^{SE}\}],$$

$$\left| \Delta b - \frac{1}{2} \Delta Q \right| + \left| \Delta \beta - \frac{1}{2} \Delta P \right| \leq \frac{1}{r} [S^{RO} - \max\{S^{SO}, S^{SE}\}].$$

The first best requires $\Delta b = \Delta Q$ and $\Delta \beta = 0$. Therefore,

$$\text{RE first-best feasible if } \Delta Q \leq \frac{1}{r} [S^{FB} - \max\{S^{SO}, S^{SE}\}],$$

$$\text{RO first-best feasible if } \frac{1}{2}(\Delta Q + \Delta P) \leq \frac{1}{r} [S^{FB} - \max\{S^{SO}, S^{SE}\}].$$

Interpretation. The same first-best incentives create different credibility requirements under employment and outsourcing.

5 Identification / theoretical strategy

5.1 Controlled theoretical comparison

The paper does not use empirical identification; it uses a formal theoretical comparison. Technology, preferences, information, and noncontractibility are held fixed. Only asset ownership and governance mode vary.

Interpretation. The identifying variation in the model is conceptual: change legal ownership of the asset while keeping the same relational promise and technology.

5.2 Separation of incentives from enforcement

A given relational contract generates the same upstream actions under employment and outsourcing. The action choice is driven by the promised bonus schedule. Feasibility differs because reneging payoffs differ.

Interpretation. This is what lets the paper say ownership matters even when marginal incentives are held constant.

5.3 Fallback governance as the enforcement threat

After reneging, the parties revert to the best spot governance arrangement. The more attractive the fallback, the harder it is to sustain a relational contract.

Interpretation. The fallback option is double-edged: it is useful outside the relationship, but it weakens the credibility of promises inside the relationship.

5.4 Comparative statics as theoretical tests

Result 1 varies ΔP to study supply-price volatility. Result 2 compares high-powered incentives under the two ownership structures. Result 3 varies the discount rate r . Result 4 varies payoff levels while holding marginal incentives fixed. Result 5 studies the existence problem created by a mediocre spot market.

Interpretation. Each result changes one force while holding others fixed. This clarifies the distinct predictions of the relational-contract view.

5.5 Threats and limits

The model abstracts from formal contracts except for asset ownership, uses a stark trigger-strategy punishment, models only two parties, and uses a linear example for many comparative results. Empirical implementation would require measuring informal bonuses, outside options, discount rates, and reneging temptations.

Interpretation. The paper is best read as a theory of mechanism and comparative statics, not as a direct empirical test.

6 Figures and interpretations

6.1 Figure I: Combinations of Ownership and Governance Regimes That Define Four Organizational Forms

Read:

- The columns distinguish asset ownership: nonintegrated/upstream ownership versus integrated/downstream ownership.
- The rows distinguish governance: spot governance versus relational governance.
- The four cells are spot outsourcing, spot employment, relational outsourcing, and relational employment.
- This taxonomy shows that the boundary of the firm and the existence of relational contracts are separate dimensions.

Interpretation. Figure I is the organizing map for the paper. Static property-rights models mainly compare the top row. Repeated-game models of employment or supply relationships often compare one column. This paper’s main contribution is to compare the bottom row directly.

Economic message. The firm boundary cannot be understood by asking only who owns assets or only whether informal contracts exist. Ownership and relational contracting jointly determine feasible organization.

Governance Environment	Ownership Environment	
	Nonintegrated Asset Ownership	Integrated Asset Ownership
	Spot	Spot Employment
Spot	Spot Outsourcing	Spot Employment
	Relational Outsourcing	Relational Employment

FIGURE I
Combinations of Ownership and Governance Regimes That Define Four Organizational Forms: Spot Outsourcing, Spot Employment, Relational Outsourcing, and Relational Employment

Figure I. Four organizational forms defined by ownership and governance regimes.

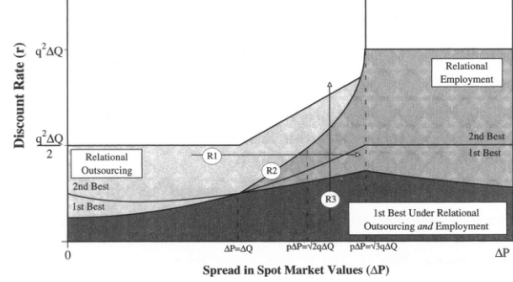


FIGURE II
Efficient Organization Form as a Function of r and ΔP
($q(a) = qa_1$, $p(a) = pa_2$, and $c(a) = \frac{1}{2}a_1^2 + \frac{1}{2}a_2^2$)
The figure shows how the efficient organizational form varies with the discount rate (r) and the difference between the high and low market valuations (ΔP), assuming that $p = q = 1$. Our first three results are illustrated in (R1) through (R3) (see the text for a complete discussion).

Figure II. Efficient organization form in the linear two-task example.

6.2 Figure II: Efficient Organization Form as a Function of r and ΔP

Read:

- The vertical axis is the discount rate r . A higher r means a lower value of future relationships, making relational contracts harder to sustain.
- The horizontal axis is ΔP , the spread in alternative-use values, which captures volatility or importance of spot-market values.
- Low r allows first-best relational governance because future surplus is large enough to deter reneging.
- Large ΔP makes relational outsourcing harder to sustain because the spot price creates larger reneging temptations.
- The labeled arrows $R1$ – $R3$ illustrate the first three results.

Interpretation. Figure II converts the abstract reneging constraints into an organizational map. The boundaries are not production frontiers; they are credibility frontiers.

Economic message. The efficient boundary of the firm changes with relational capital and outside-option volatility. A transaction may move from outsourcing to integration because integration improves the credibility of informal governance.

7 Interpretive synthesis

- **Why ownership matters.** Ownership determines who controls the good after reneging and changes spot bargaining benchmarks.

- **Why relational contracts matter.** Without relational contracts, spot employment creates no upstream effort and spot outsourcing can distort effort toward outside value.
- **Why integration can help.** Integration can remove the sensitivity of renegeing temptations to volatile alternative-use values.
- **Why integration can hurt.** Inside the firm, high-powered bonuses are pure promised transfers; if they are too large, the downstream owner wants to renege.
- **Why markets can block firms.** If spot outsourcing is a credible fallback, the value of continuing employment may be too low to sustain a relational-employment improvement.
- **Why managers matter.** Managers create value by maintaining the credibility of informal agreements and by revising them without destroying trust.

8 Short summary

- The paper studies how relational contracts interact with asset ownership.
- There is an upstream party, downstream party, and asset used to produce a good.
- The good has downstream value Q and alternative-use value P .
- Actions are costly and unobservable; Q and P are observable but not court-verifiable.
- Spot outsourcing gives upstream bargaining incentives but may distort effort toward outside value.
- Spot employment gives no noncontractible effort in the baseline model.
- Relational employment and relational outsourcing can use informal bonuses based on Q and P .
- The same bonus schedule induces the same actions under both ownership structures.
- But the renegeing temptation differs because the spot fallback differs.
- Integration can make some relational contracts feasible and others infeasible.
- The model formalizes why firms cannot simply mimic markets internally.
- Vertical integration can be efficient when supply values are volatile.
- Performance payments tend to be weaker inside firms than between firms.
- The optimal firm boundary can depend on discount rates and payoff levels, not only marginal incentives.
- A mediocre spot market can prevent the existence of a more efficient firm by weakening relational enforcement.

9 Conclusion

9.1 Core conclusion

Baker, Gibbons, and Murphy show that a theory of the firm must account for both formal ownership and informal relational contracting. Asset ownership changes renegeing temptations, which changes which relational contracts are feasible.

9.2 Incremental contribution

The paper makes the boundary of the firm an enforcement instrument for relational contracts. It goes beyond Grossman–Hart by showing that ownership affects the credibility of informal promises, and goes beyond standard repeated-game models by showing that whether a relationship is inside or between firms matters for self-enforcement.

9.3 Final takeaway

Firms, suppliers, alliances, boards, and internal divisions are governed by informal promises whose credibility depends on formal structure. Ownership, hierarchy, and management matter because they shape the fallback payoffs that discipline those promises.